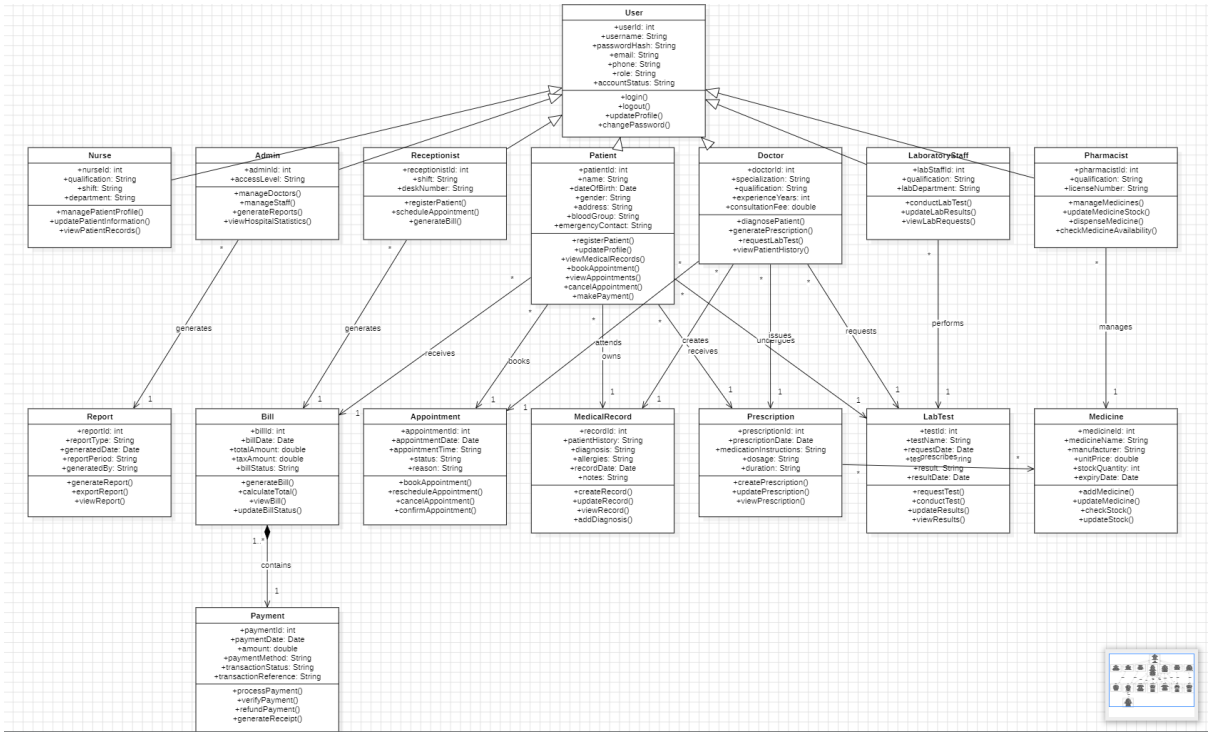
UML Class Diagram Design

1. Diagram



2. Identification of Classes

The following classes are derived from the use cases in the provided diagram.

Class	Purpose
User	Stores common login and authentication information.
Patient	Maintains patient profiles and medical records and handles appointments and payments.
Doctor	Diagnoses patients, generates prescriptions, and requests laboratory tests.
Nurse	Manages patient profiles and assists in patient care
Admin	Manages doctors, staff, and hospital reports.
Receptionist	Registers patients and generates bills.
Laboratory Staff	Conducts laboratory tests and updates laboratory results.
Pharmacist	Manages medicines and medication inventory.
Appointment	Stores appointment details, status, and scheduling information.
Medical Record	Stores diagnoses, patient history, and medical information.
Prescription	Stores prescribed medicines and dosage instructions.
Lab Test	Stores test requests, test details, and laboratory results.

Bill	Stores billing details and the total amount payable.
Payment	Records payment transactions and their status.
Medicine	Maintains medicine names, stock, prices, and expiry dates.
Report	Represents reports generated for hospital administration.

3. Attributes and Methods of Each Class

3.1 User Class

Attributes: userId, username, passwordHash, email, phone, role, accountStatus

Methods:

- login()
- logout()
- updateProfile()
- changePassword()

3.2 Patient Class

Attributes: patientId, name, dateOfBirth, gender, address, bloodGroup, emergencyContact

Methods:

- registerPatient()
- updateProfile()
- viewMedicalRecords()
- bookAppointment()
- viewAppointments()
- cancelAppointment()
- makePayment()

3.3 Doctor Class

Attributes: doctorId, specialization, qualification, experienceYears, consultationFee

Methods:

- diagnosePatient()
- generatePrescription()
- requestLabTest()
- viewPatientHistory()

3.4 Nurse Class

Attributes: nurseId, qualification, shift, department

Methods:

- managePatientProfile()
- updatePatientInformation()
- viewPatientRecords()

3.5 Admin Class

Attributes: adminId, accessLevel

Methods:

- manageDoctors()
- manageStaff()
- generateReports()
- viewHospitalStatistics()

3.6 Receptionist Class

Attributes: receptionistId, shift, deskNumber

Methods:

- registerPatient()
- scheduleAppointment()
- generateBill()

3.7 LaboratoryStaff Class

Attributes: labStaffId, qualification, labDepartment

Methods:

- conductLabTest()
- updateLabResults()
- viewLabRequests()

3.8 Pharmacist Class

Attributes: pharmacistId, qualification, licenseNumber

Methods:

- manageMedicines()
- updateMedicineStock()
- dispenseMedicine()
- checkMedicineAvailability()

3.9 Appointment Class

Attributes: appointmentId, appointmentDate, appointmentTime, status, reason

Methods:

- bookAppointment()
- rescheduleAppointment()
- cancelAppointment()
- confirmAppointment()

3.10 MedicalRecord Class

Attributes: recordId, patientHistory, diagnosis, allergies, recordDate, notes

Methods:

- createRecord()
- updateRecord()
- viewRecord()
- addDiagnosis()

3.11 Prescription Class

Attributes: prescriptionId, prescriptionDate, medicationInstructions, dosage, duration

Methods:

- createPrescription()
- updatePrescription()
- viewPrescription()

3.12 LabTest Class

Attributes: testId, testName, requestDate, testStatus, result, resultDate

Methods:

- requestTest()
- conductTest()
- updateResults()
- viewResults()

3.13 Bill Class

Attributes: billId, billDate, totalAmount, taxAmount, billStatus

Methods:

- generateBill()
- calculateTotal()

- viewBill()
- updateBillStatus()

3.14 Payment Class

Attributes: paymentId, paymentDate, amount, paymentMethod, transactionStatus, transactionReference

Methods:

- processPayment()
- verifyPayment()
- refundPayment()
- generateReceipt()

3.15 Medicine Class

Attributes: medicineId, medicineName, manufacturer, unitPrice, stockQuantity, expiryDate

Methods:

- addMedicine()
- updateMedicine()
- checkStock()
- updateStock()

3.16 Report Class

Attributes: reportId, reportType, generatedDate, reportPeriod, generatedBy

Methods:

- generateReport()
- exportReport()
- viewReport()